

Chemical Equilibrium

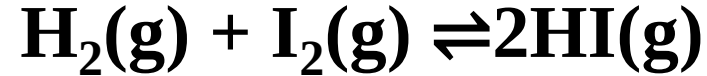
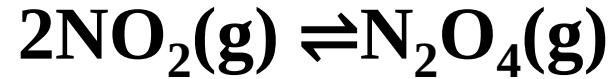
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Reversible Reaction

A reaction which can go in the forward and backward direction simultaneously is called a Reversible reaction.



Difference between Reversible and Irreversible Process

Reversible process	Irreversible process
1.The process that can be reversed in order to obtain the original state of a system are called Reversible Processes.	1.The process that can not be reversed to get back the original state of a system is called Irreversible process.
2.Can be reversed	2.Cannot be reversed
3.Changes that occur is infinite in the system	3.Changes that occur is finite in the system.
4.There is an equilibrium between the initial and final state of the system.	4.There is no equilibrium in the system.
5.Maximum work is obtained in the process.	5.The work we get is not maximum.
6.Almost time taken to complete this reaction is infinity.	6.Time taken for the completion of the reaction is finite.

❖ How a reversible reaction can be converted to irreversible reaction?

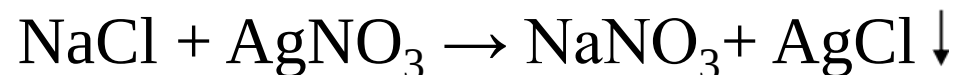
1. When there is an escape of the gas products through an open system: In an open vessel, calcium carbonate decomposes into calcium oxide and carbon dioxide. During the reaction, the carbon dioxide can be removed from the vessel so, only the forward reaction is possible as the concentration of the reactants is reduced due to the escape of carbon dioxide. Therefore, there is no equilibrium attained during the reaction.



In a closed vessel, calcium carbonate decomposes into calcium oxide and carbon dioxide. During the reaction, the carbon dioxide cannot be removed from the vessel so, the reverse reaction is also possible. Therefore, there is an equilibrium during the reaction.

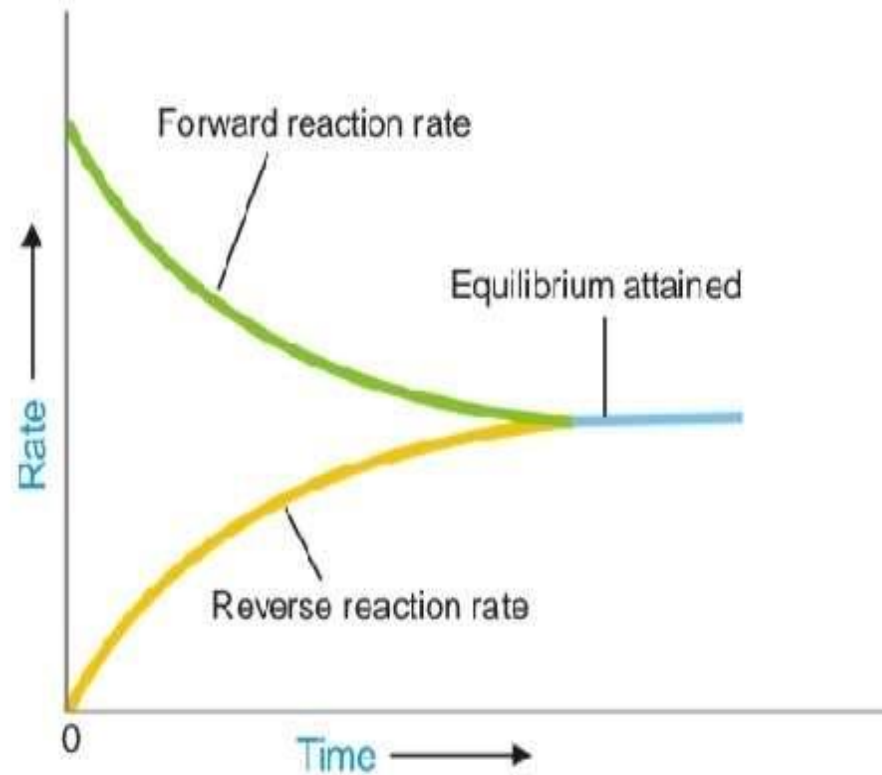


2. By removing the product: By removing the product it is possible to make the reversible reaction irreversible as this condition will always shift the equation in the forward direction.



Chemical Equilibrium

The state of a reversible reaction when the two opposing reactions occur at the same rate and the concentrations of reactants and products do not change with time.



At equilibrium the forward reaction rate equals the reverse reaction rate.

Chemical Equilibrium is Dynamic Equilibrium

Chemical equilibrium is achieved when the forward reaction rates and backward reaction rates are same. So although overall it is in equilibrium but inside the forward reaction and backward reaction both occur and thus there is motion in the atomic and molecular level. And dynamic refers to anything having motion.

Since the equilibrium is not static so termed as dynamic equilibrium.

Charecteristics of Chemical Equilibrium

The main characteristics of chemical equilibrium are as follows:

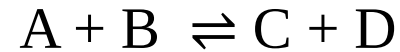
- At chemical equilibrium, the rate of forward and reverse reactions are equal.
- Equilibrium cannot be attained in an Open Vessel
- All the reactants and products are present at equilibrium.
- The position of equilibrium is not affected by a catalyst.
- Value of Equilibrium Constant does not depend upon the initial concentration of reactants.
- The concentrations of the reactants and products reach constant values at equilibrium.
- At equilibrium the Gibbs free energy (G) is minimum and any change taking place at equilibrium proceeds without change in free energy i.e. $\Delta G = 0$.

Law of Mass Action

Two Norwegian chemists, Guldberg and Waage, studied experimentally a large number of equilibrium reactions. In 1864, they postulated a generalisation called the Law of Mass action. It states that : the rate of a chemical reaction is proportional to the active masses of the reactants.

By the term ‘active mass’ is meant the molar concentration i.e., number of moles per litre. It is expressed by enclosing the formula of the substance in square brackets.

Let us consider a general reaction



and let [A], [B], [C] and [D] represent the molar concentrations of A, B, C and D at the equilibrium point. According to the Law of Mass action.

Rate of forward reaction $\propto [A] [B] = k_1 [A] [B]$

Rate of reverse reaction $\propto [C][D] = k_2 [C][D]$

where k_1 and k_2 are rate constants for the forward and reverse reactions.

At equilibrium, rate of forward reaction = rate of reverse reaction.

Therefore $k_1[A] [B] = k_2 [C] [D]$

$$\frac{k_1}{k_2} = \frac{[C][D]}{[A][B]}$$

At any specific temperature k_1/k_2 is constant since both k_1 and k_2 are constants. The ratio k_1/k_2 is called Equilibrium constant and is represented by the symbol K_c Or simply k . The subscript 'c' indicates that the value is in terms of concentrations of reactants and products. The equation (1) may be written as

$$\text{Equilibrium constant} \quad k_c = \frac{[C][D]}{[A][B]}$$

— Product concentrations
 — Reactant concentrations

This equation is known as the Equilibrium constant expression or Equilibrium law.

Other Characteristics of K_c

- 1) Equilibrium can be approached from either direction.
- 2) K_c does not depend on the initial concentrations of reactants and products.
- 3) K_c does depend on temperature.

Magnitude of K_c

- ⇒ If the K_c value is large ($K_c \gg 1$), the equilibrium lies to the right and the reaction mixture contains mostly products.
- ⇒ If the K_c value is small ($K_c \ll 1$), the equilibrium lies to the left and the reaction mixture contains mostly reactants.
- ⇒ If the K_c value is close to 1 ($0.10 < K_c < 10$), the mixture contains appreciable amounts of both reactants and products.

Relation Between K_c and K_p

Let us consider a general reaction



where all reactants and products are gases. We can write the equilibrium constant expression in terms of partial pressures as

$$K_p = \frac{(p_C)^l (p_D)^m}{(p_A)^j (p_B)^k} \quad \dots(1)$$

Assuming that all these gases constituting the equilibrium mixture obey the ideal gas equation, the partial pressure (p) of a gas is

$$p = \left(\frac{n}{V} \right) RT$$

Where $\frac{n}{V}$ is the molar concentration. Thus the partial pressures of individual gases, A, B, C and D are:

$$P_A = [A] RT; \quad p_B = [B] RT; \quad p_C = [C] RT; \quad p_D = [D] RT$$

The Equilibrium Constant in Terms Of Partial Pressures

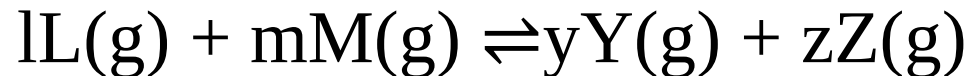
When all the reactants and products are gases, we can also formulate the equilibrium constant expression in terms of partial pressure. The relationship between the partial pressure (p) of any one gas in the equilibrium mixture and the molar concentration follows from the general ideal gas equation

$$pV = nRT \text{ or, } p = (n/V) RT$$

The quantity (n/V) is the number of moles of the gas per unit volume and is simply the molar concentration.

$$\text{Thus, } p = (\text{Molar concentration}) \times RT$$

i.e., the partial pressure of a gas in the equilibrium mixture is directly proportional to its molar concentration at a given temperature



$$K_p = \frac{(p_Y)^y (p_Z)^z}{(p_L)^l (p_M)^m}$$

Substituting these values in equation (1), we have

$$K_p = \frac{[C]^l (RT)^l [D]^m (RT)^m}{[A]^j (RT)^j [B]^k (RT)^k}$$

or

$$K_p = \frac{[C]^l [D]^m}{[A]^j [B]^k} \times \frac{(RT)^{l+m}}{(RT)^{j+k}}$$

$$K_p = K_c \times (RT)^{(l+m)-(j+k)}$$

$$\therefore K_p = K_c \times (RT)^{\Delta n} \quad \dots(2)$$

where $\Delta n = (l+m) - (j+k)$, the difference in the sums of the coefficients for the gaseous products and reactants.

From the expression (2) it is clear that when $\Delta n = 0$, $K_p = K_c$.

- When $\Delta n = +V_e$, the number of moles of reactant are less than the number of moles of products, thus $K_p > K_c$. $\square$
- When $\Delta n = -V_e$, the number of moles of reactant are more than the number of moles of products, thus $K_p < K_c$.
- When $\Delta n = 0$, the number of moles of reactant equals number of moles of product, thus $K_p = K_c$

1. At 500°C , the reaction between N_2 and H_2 to form ammonia has $K_c = 6.0 \times 10^{-2}$. What is the numerical value of K_p for the reaction?

Here, we will use the general expression

$$K_p = K_c (RT)^{\Delta n}$$

For the reaction $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$

we have $\Delta n = (\text{sum of quotients of products}) - (\text{sum of quotients of reactants})$
 $= 2 - 4 = -2$

$K_c = 6.0 \times 10^{-2}$, $T = 500 + 273 = 773 \text{ K}$, $R = 0.0821$

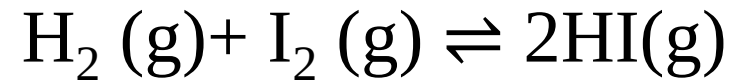
Substituting the value of R , T , K_c and Δn in the general expression, we have

$$K_p = (6.0 \times 10^{-2}) [(0.0821) \times (773)]^{-2}$$

$$K_p = 1.5 \times 10^{-5}$$

Calculation of concentration of reactants and products at equilibrium

Let us consider the formation of HI in which, 'a' moles of hydrogen and 'b' moles of iodine gas are allowed to react in a container of volume V. Let 'x' moles of each of H₂ and I₂ react together to form 2x moles of HI



Initial no. of moles	a	b	0
No. of moles reacted	x	x	0
No. of moles at eq.	a-x	b-x	2x
Active mass or molar coc. at eq.	a-x/v	b-x/v	2x/v

Applying law of mass action,

$$K_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]}$$
$$= \frac{\left(\frac{2x}{V}\right)^2}{\left(\frac{a-x}{V}\right)\left(\frac{b-x}{V}\right)} = \frac{4x^2}{(a-x)(b-x)}$$

Dissociation of PCl_5 :

Consider that ‘a’ moles of PCl_5 is taken in a container of volume V. Let ‘x’ moles of PCl_5 be dissociated into x moles of PCl_3 and x moles of Cl_2 .

	$\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$		
Initial no. of moles	a	0	0
No. of moles dissociated	x	0	0
No. of moles at eq.	a-x	x	x
Active mass or molar conc. at equilibrium	a-x/v	x/v	x/v

Applying law of mass action,

$$K_c = \frac{[\text{PCl}_3][\text{Cl}_2]}{[\text{PCl}_5]} = \frac{\left(\frac{x}{V}\right)\left(\frac{x}{V}\right)}{\left(\frac{a-x}{V}\right)} = \frac{x^2}{(a-x)V}$$

Synthesis of ammonia:

Let us consider the formation of ammonia in which, ‘a’ moles nitrogen and ‘b’ moles hydrogen gas are allowed to react in a container of volume V. Let ‘x’ moles of nitrogen react with 3x moles of hydrogen to give 2x moles of ammonia.



Initial no. of moles	a	b	0
No. of moles reacted	x	3x	2x
No. of moles at eq.	a-x	b-3x	2x
Active mass or molar conc. at eq.	a-x/v	b-3x/v	2x/v

Applying law of mass action,

$$K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$$

$$= \frac{\left(\frac{2x}{V}\right)^2}{\left(\frac{a-x}{V}\right)\left(\frac{b-3x}{V}\right)^3}$$

$$= \frac{\left(\frac{4x^2}{V^2}\right)}{\left(\frac{a-x}{V}\right)\left(\frac{b-3x}{V}\right)^3}$$

$$K_c = \frac{4x^2 V^2}{(a-x)(b-3x)^2}$$

❖ In the dissociation of PCl_5 as: $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ if the degree of dissociation is x at equilibrium pressure P , then the equilibrium constant for the reaction is:

	$\text{PCl}_5 (\text{g})$	$\rightleftharpoons$	$\text{PCl}_3 (\text{g})$	$+$	$\text{Cl}_2 (\text{g})$
Initially	a		0		0
At equilibrium	$(a-x)$		x		x
Active mass	$\frac{(a-x)}{v}$		$\frac{x}{v}$		$\frac{x}{v}$

$$K_p = \frac{p_{\text{PCl}_3} \times p_{\text{Cl}_2}}{p_{\text{PCl}_5}} = \frac{\left(\frac{x}{a+x} \cdot P\right) \times \left(\frac{x}{a+x} \cdot P\right)}{\left(\frac{a-x}{a+x}\right) P} = \frac{x^2 P}{(a+x)(a-x)} = \frac{x^2 P}{a^2 - x^2}$$

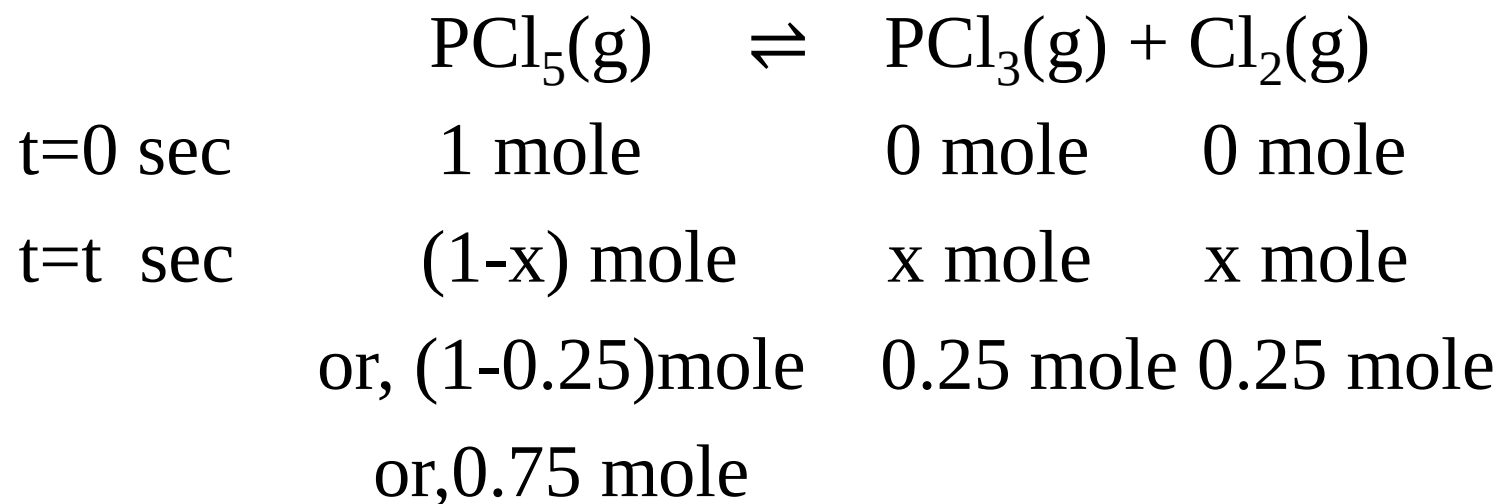
$$K_c = \frac{[\text{PCl}_3][\text{Cl}_2]}{[\text{PCl}_5]} = \frac{\frac{x}{v} \times \frac{x}{v}}{\frac{(a-x)}{v}} = \frac{x^2}{(a-x)v}$$

Total number of moles $= a - x + x + x = a + x$

$P_{\text{PCl}_5} = (a-x/a+x) \times P$, $P_{\text{PCl}_3} = (x/a+x) \times P$, $P_{\text{Cl}_2} = (x/a+x) \times P$

2. At 70°C and 1.75 atm pressure 25% PCl_5 dissociates to PCl_3 and Cl_2 . Calculate K_p and K_c for the reaction.

Solution:



Total mole = $0.75 + 0.25 + 0.25 = 1.25$ mole

$$P_{\text{PCl}_5} = 1.75 \times 0.75/1.25 = 1.05 \text{ atm}$$

$$P_{\text{PCl}_3} = 1.75 \times 0.25/1.25 = 0.35 \text{ atm}$$

$$P_{\text{Cl}_2} = 1.75 \times 0.25/1.25 = 0.35 \text{ atm}$$

$$\begin{aligned} K_p &= x^2 \times p / a^2 - x^2 \\ &= 0.116 \text{ atm} \end{aligned}$$

$$K_p = K_c(RT)^{\Delta n}$$

$$K_c = K_p/RT$$

$$= 0.116 / 0.0821 \times 343$$

$$= 4.14 \times 10^{-3} \text{ mol/L}$$

3. K_p for a reaction PCl_5 gives $\text{PCl}_3 + \text{Cl}_2$ at 250°C is 0.82. Calculate the degree of dissociation at given temperature under a total pressure of 5 atm.



$$P_{\text{PCl}_5} = \frac{1 - \alpha}{1 + \alpha} P_{\text{Total}}$$

$$P_{\text{PCl}_3} = \frac{\alpha}{1 + \alpha} P_{\text{Total}}$$

$$P_{\text{PCl}_2} = \frac{\alpha}{1 + \alpha} P_{\text{Total}}$$

$$K_p = \frac{(P_{\text{PCl}_3})(P_{\text{Cl}_2})}{(P_{\text{PCl}_5})} = \frac{\alpha^2 P_{\text{Total}}}{1 - \alpha^2}$$

$$0.82 = \frac{\alpha^2 \times 5}{1 - \alpha^2}$$

$$\frac{\alpha^2}{1 - \alpha^2} = \frac{0.82}{5} = 0.164$$

$$\alpha^2 = 0.164 - 0.164\alpha^2$$

$$1.16\alpha^2 = 0.164\alpha^2$$

$$1.164\alpha^2 = 0.164$$

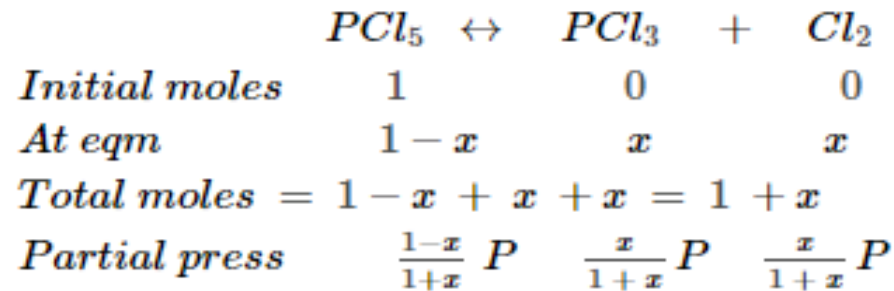
$$\alpha^2 = \frac{0.164}{1.164} = 0.14$$

$$\alpha = 0.37$$

$$37\%$$

4. PCl_5 is 50% dissociated into PCl_3 and Cl_2 at 1 atmosphere. What pressure it will be 40% dissociated?

The given decomposition reaction is



$$\begin{aligned} K_P &= \frac{P_{\text{PCl}_3} \times P_{\text{Cl}_2}}{P_{\text{PCl}_5}} \\ &= \frac{\frac{x}{1+x} P \frac{x}{1+x} P}{\frac{1-x}{1+x} P} = \frac{x^2 P}{1^2 - x^2} \\ &= \frac{(0.5)^2 \times 1}{1 - (0.5)^2} = \frac{1}{3} \quad \left(\text{since } x = 50 \% \text{ dissociation} \right) \end{aligned}$$

For 40% dissociation

$$K_P = \frac{x^2 P}{1^2 - x^2} \quad .$$

$$\frac{1}{3} = \frac{(0.4)^2 \times P}{1 - (0.4)^2}$$

$$P = \frac{0.84}{0.16 \times 3} = 1.75 \text{ atm}$$

The pressure required for 40 % dissociation = 1.75 atm

5. At certain temperature and under a pressure of 4 atm, PCl_5 is 10% dissociated. Calculate the pressure at which PCl_5 will be 20% dissociated at temperature remaining constant.



Total no. of moles in the equilibrium mixture = $1 - \alpha + \alpha + \alpha = (1 + \alpha)$ mol.

Let the total pressure of equilibrium mixture = p atm

$$\text{Partial pressure of PCl}_5, P_{\text{PCl}_5} = \frac{1-\alpha}{1+\alpha} \times p \text{ atm}$$

$$\text{Partial pressure of PCl}_3 = \frac{\alpha}{1+\alpha} \times p \text{ atm}$$

$$\text{Partial pressure of Cl}_2, P_{\text{Cl}_2} = \frac{\alpha}{1+\alpha} \times p \text{ atm}$$

$$\begin{aligned} K_p &= \frac{P_{\text{PCl}_3} \times P_{\text{Cl}_2}}{P_{\text{PCl}_5}} \\ &= \frac{\left(\frac{\alpha}{1+\alpha} p \text{ atm} \right) \times \left(\frac{\alpha}{1+\alpha} p \text{ atm} \right)}{\frac{1-\alpha}{1+\alpha} p \text{ atm}} \\ &= \frac{\alpha^2 p}{1-\alpha^2} \end{aligned}$$

$$P = 4 \text{ atm and } \alpha = 10\% = \frac{10}{100} = 0.1$$

$$\begin{aligned} K_p &= \frac{(0.1) \times (0.1) \times (4 \text{ atm})}{1 - (0.1)^2} \\ &= \frac{0.04}{0.99} \\ &= 0.04 \text{ atm} \end{aligned}$$

calculation of P under new condition

$$\alpha = 0.2, K_p = 0.04 \text{ atm}$$

$$K_p = \frac{\alpha^2 p}{1 - \alpha^2} \text{ or } p = \frac{K_p(1 - \alpha^2)}{\alpha^2}$$

$$= \frac{(0.04 \text{ atm}) [(1 - (0.2)^2)]}{(0.2)^2}$$

$$= \frac{0.04 \text{ atm} \times 0.96}{0.04}$$

$$= 0.96 \text{ atm}$$

Heterogeneous Equilibria

The equilibrium which the reactants and products are not all in the same phase are called heterogeneous equilibria.

If the reaction is carried in a closed vessel, the following equilibrium is established.



Straight forward application of Equilibrium law to the decomposition reaction of calcium carbonate leads to the equilibrium constant expression.

$$K = \frac{[\text{CO}_2][\text{CaO}]}{[\text{CaCO}_3]}$$

But CaCO_3 and CaO are pure solids. The concentration (moles per unit volume) of a pure solid (or liquid) is fixed and cannot vary. Thus **the concentrations of pure solids or liquids are not included in the equilibrium constant expression.**

Ignoring the concentrations of CaCO_3 and CaO , the equilibrium constant expression for the decomposition of CaCO_3 may be written as

$$K_c = [\text{CO}_2]$$

In terms of partial pressures

$$K_p = p_{\text{CO}_2}$$

Similarly, the equilibrium constant expression for the decomposition of liquid water



would be

$$K = \frac{[\text{H}_2]^2 [\text{O}_2]}{[\text{H}_2\text{O}]}$$

Ignoring the concentration of liquid water, we have

$$K_c = [\text{H}_2]^2 [\text{O}_2]$$

$$K_p = (p_{\text{H}_2})^2 (p_{\text{O}_2})$$

Le Chatelier's principle

At the equilibrium of reversible reaction if any of the factors (Temperature, Pressure and concentration of reactant) is changed, the equilibrium position will shift in such a way that the effect of change of factor is relieved.

There are three ways in which the stress can be caused on a chemical equilibrium:

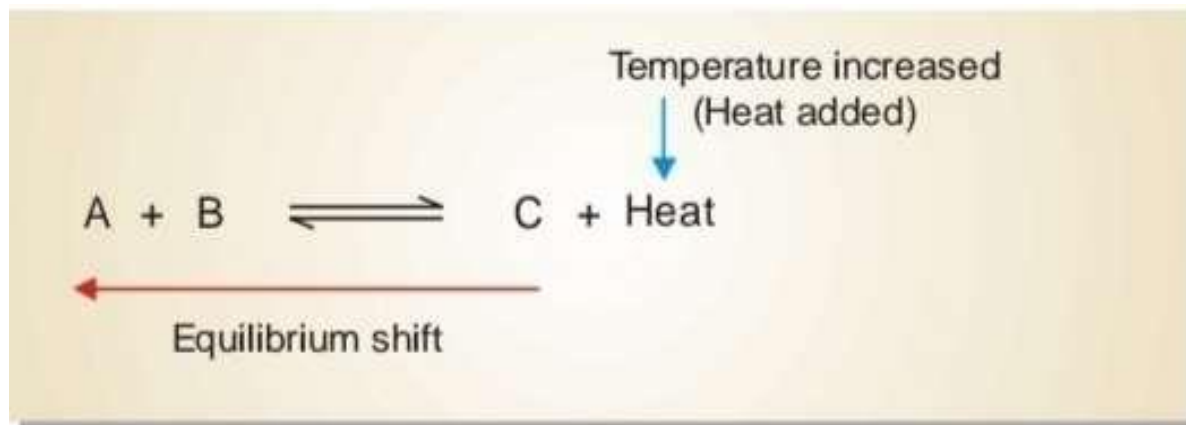
- (1) Changing the concentration of a reactant or product.
- (2) Changing the pressure (or volume) of the system.
- (3) Changing the temperature

Effect of Change of Temperature

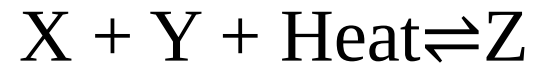
Let us consider an **exothermic reaction**



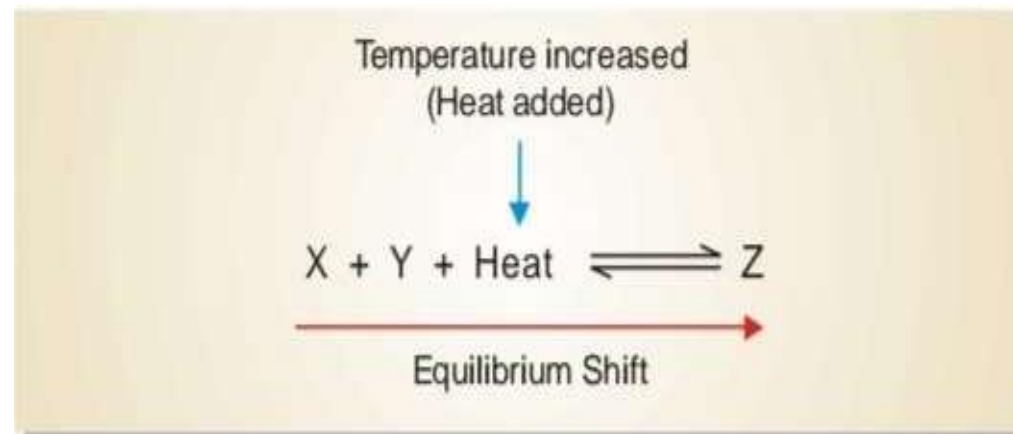
When the temperature of the system is increased, heat is supplied to it from outside. According to Le Chatelier's principle, the equilibrium will shift to the left which involves the absorption of heat. This would result in the increase of the concentration of the reactants A and B.



In an endothermic reaction



the increase of temperature will shift the equilibrium to the right as it involves the absorption of heat. This increases the concentration of the product.

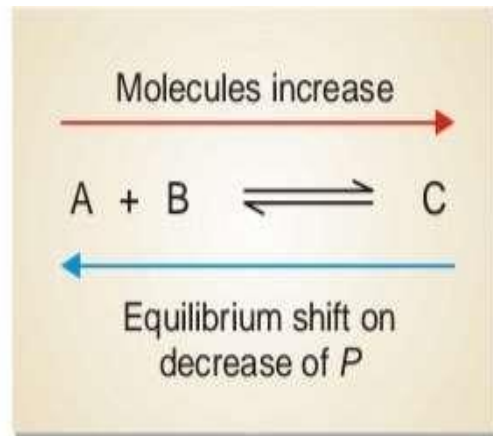
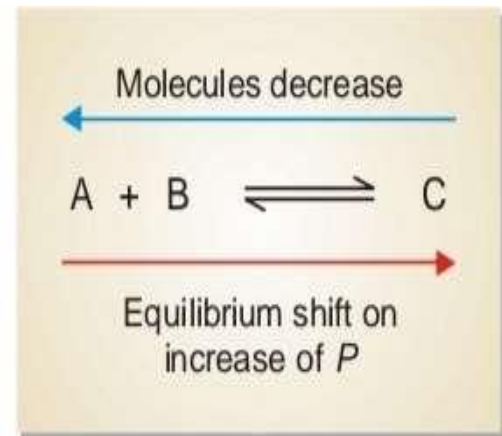


Effect of a Change in Pressure

Let us consider a reaction,



The combination of A and B produces a decrease of number of molecules while the decomposition of C into A and B results in the increase of molecules. Therefore, by the increase of pressure on the equilibrium it will shift to right and give more C. A decrease in pressure will cause the opposite effect. The equilibrium will shift to the left when C will decompose to form more of A and B.



System	Pressure increased	Pressure decreased
(1) $N_2O_4 \rightleftharpoons 2NO_2$	←	→
(2) $PCl_5 \rightleftharpoons PCl_3 + Cl_2$	←	→
(3) $2SO_2 + O_2 \rightleftharpoons 2SO_3$	→	←
(4) $N_2 + 3H_2 \rightleftharpoons 2NH_3$	→	←
(5) $H_2 + I_2 \rightleftharpoons 2HI$	No effect	No effect
(6) $N_2 + O_2 \rightleftharpoons 2NO$	No effect	No effect

Effect of a Change in Concentration

Let us consider a general reaction



When a reactant, say, A is added at equilibrium, its concentration is increased. The forward reaction alone occurs momentarily. According to Le Chatelier's principle, a new equilibrium will be established so as to reduce the concentration of A. Thus the addition of A causes the equilibrium to shift to right. This increases the concentration (yield) of the product C.

Following the same line of argument, a decrease in the concentration of A by its removal from the equilibrium mixture, will be undone by shift to the equilibrium position to the left. This reduces the concentration (yield) of the product C.

